

Coregonus artedii

Cisco



Length Up to 22½ in (57 cm)
Weight Up to 7¼ lb (3.5 kg)
Sex Male/Female
Status Least concern

Location N. North America

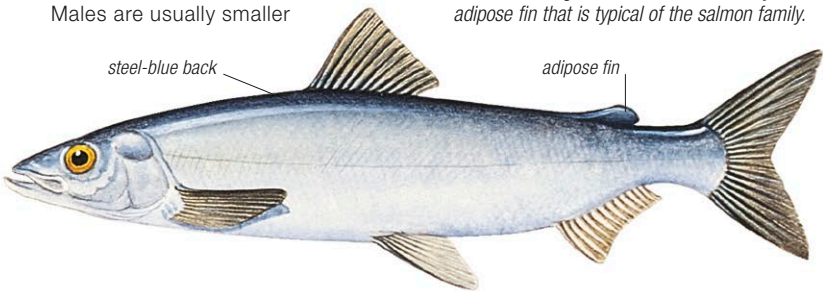


Essentially a freshwater fish, the cisco, or lake herring, may also be found in brackish water and very occasionally in salt water. It is usually found in large shoals in mid-water, the depth varying with the season. Basically a plankton-feeder, it will also consume a wide variety of other foods. Spawning occurs in autumn and is dependent on the temperature of water. Males are usually smaller

than the females and, during the breeding season, the males develop small bumps (pearl organs) that function to stimulate the female to lay her eggs. The young feed on algae and plankton, whereas adults are more predatory and also feed on insects and small crustaceans. The cisco differs from other *Coregonus* species in having upper and lower jaws of the same length and more gill rakers. A major threat to this species is the introduction to lakes of the alewife and lamprey. However, it is also sensitive to pollution and the recent rapid decline of *Diporeia*, a tiny crustacean that is a major food source to lake fishes.

OPEN-WATER FISH

This slender, silvery fish has the characteristic dark upper and light lower body of an open-water schooling fish. It also has the fleshy adipose fin that is typical of the salmon family.



Salmo trutta

Brown trout



Length Up to 4½ ft (1.4 m)
Weight Up to 33 lb (15 kg)
Sex Male/Female
Status Least concern

Location Northern Europe; introduced worldwide



Thanks to widespread introductions, the brown trout can be found in clean, cool, freshwater rivers, streams, lakes, and stocked ponds worldwide, but it is native to northern Europe and western Asia. While most individuals spend their whole lives in fresh water, some populations, known as sea trout,

migrate down into coastal waters after 2 to 3 years as silvery “smolts,” later returning to fresh water to breed. Populations of brown trout are highly variable in their background color and in the number and size of their characteristic spots. Large red and black spots, haloed with white, are common in adults, which have a large mouth extending well beyond the eye.



Thaleichthys pacificus

Eulachon



Length Up to 11½ in (30 cm)
Weight Up to 2½ oz (60 g)
Sex Male/Female
Status Least concern

Location E. North Pacific, W. North America



This is a member of the smelt family—a group of slender, silvery fishes that also includes the capelin (see opposite). Like the capelin, it has a long body

and underslung lower jaw, but is blue to blue-brown above. Adult eulachons live at sea, where they feed on plankton, filtering their food from the water. However, they travel upstream in medium- to large-sized rivers to spawn. Few survive spawning, which limits the fish’s lifespan to between 2 and 4 years. Eulachons are both a sport and a subsistence resource, and when caught in the mouth of rivers are so oily that they can be used as wicks for candles; hence their common name “candlefish.” Their oil used to be traded by native North Americans; the trade routes were often called “grease trails.”

Thymallus thymallus

European grayling



Length Up to 23½ in (60 cm)
Weight Up to 14 lb (6.5 kg)
Sex Male/Female
Status Least concern

Location W., C., and N. Europe



This graceful salmonid, sometimes called the “lady of the stream”, is a denizen of fast-flowing rivers and

streams, but is also found in some cool lakes. Males especially have an unusually long and tall dorsal fin, which they hold erect during spawning displays to attract females. Grayling feed on a wide variety of swimming and bottom-living invertebrates, but also take insects from the surface; their sporting qualities are appreciated by fly fishermen. Graylings’ edible flesh smells of the herb thyme. The grayling is a useful indicator of water quality because it cannot tolerate organic pollution nor temperatures above 68° F (20° C). This means it has disappeared from suitable stretches of river in some regions.



Opisthoproctus soleatus

Barrel-eye

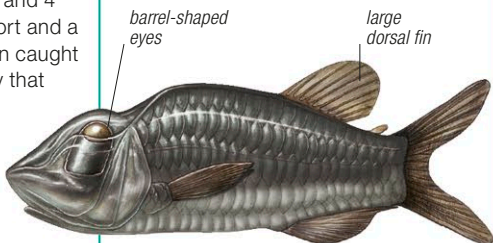


Length Up to 4¼ in (10.5 cm)
Weight Not recorded
Sex Male/Female
Status Least concern

Location Tropical and subtropical waters worldwide



Barrel-eyes are relatively small fishes named for their tubular, upward-facing eyes, an adaptation that probably helps them hunt other fishes from below. They belong to the same family as the spookfish (see right) and, like them, have most of their fins—apart from the pectoral pair—set close to the tail. This fish is dark above and silvery below, and has a luminescent organ on its rectum that contains light-producing bacteria. The light shines through a lens, and is reflected by the fish’s flattened underside, or “sole,” which spreads along the length of its body. This diffused light matches the dim light from the surface, making the barrel-eye practically invisible from below. Barrel-eyes are found in tropical waters, at depths between about 984–2,650 ft (300–800 m).



Bathylagus niger

Deep-sea smelt



Length 5½ in (14 cm)
Weight Not recorded
Sex Male/Female
Status Not evaluated

Location Southern Ocean



Found between depths of 656–5,118 ft (200–1560 m), this deep-sea smelt was discovered in the Scotia Sea between South America and Antarctica. Its habits are little known but it probably feeds on krill—which makes up nearly 50 percent of the food eaten by related Antarctic species. All *Bathylagus* smelts are small fishes of polar waters.

Dolichopteryx parini

Winged spookfish



Length 8½ in (22 cm)
Weight Not recorded
Sex Male/Female
Status Not evaluated

Location North Pacific



Belonging to a family of remarkable “barrel-eyed” fish, this fish was described in 2001 on the basis of a single specimen from the waters of the North Pacific and is named for its winglike fins. Since then other specimens have been caught, but—like other members of the genus—it remains rare. It feeds on invertebrate planktonic animals, such as free-swimming crustaceans.